HRI Survey- QRCode

Start of Block: NTU Consent Form

Human Perception of Robot Behaviour NTU-IRB ref no. IRB-2025-765 You are invited to participate in a research study on human perception of a robot's mental state from its expressive movements. This study is conducted by Bernhard Johannes Schmitt, Associate Professor, School of Art, Design and Media from Nanyang Technological University. This study will take approximately 10-12 minutes of your time. In this online survey, you will view a series of videos where a single arm robot will perform different expressive movements associated with different mental states. The questions will pertain to your recognition of the type and degree of the robot's mental state. 100 participants (location restricted to Singapore) will be recruited for this research. Your participation in this study is completely voluntary, and you have the right to withdraw from the study at any time without any penalty. If you do not wish to complete this survey, just close your browser. Your participation in this research will be kept confidential, and data will be averaged and reported in aggregate. Your responses will be anonymous and IP addresses will not be collected to guarantee complete anonymity. Possible outlet of dissemination may be research articles. Although your participation in this research may not benefit you personally, it will help us to incorporate seamless nonverbal communication between humans and robots, thereby improving human-robot interaction in healthcare, education, manufacturing, entertainment and other such industries. There will be no compensation provided. There are no risks to individuals participating in this survey beyond those that exist in daily life. If you have questions about this project, you may contact Bernhard Johannes Schmitt (+6591134078 and bjschmitt@ntu.edu.sg) This project has been reviewed and approved by NTU-Institutional Review Board. Questions concerning your rights as a participant in this research may be directed to the NTU-IRB at IRB@ntu.edu.sg or call 6592 2495. Please print a copy of this consent form for your records, if you so desire.

Consent Form

○ I have read and understood the above consent form. I certify that I am 18 years old or older and, by clicking the next button to enter the survey, I indicate my willingness to voluntarily take part in the study. (1)

○ I do not wish to participate in this study and end the survey by clicking on the next button. (2)

End of Block: NTU Consent Form

Start of Block: End of Survey if consent rejected

Thank you for your time. You have chosen not to participate in this study. Your response has not been recorded, and you may now close this window.

End of Block: End of Survey if consent rejected

Start of Block: Background Information

Section 1: Demographic Information

What is your gender?

- ○ Male (1)
- ○ Female (2)
- ○ Other (3)
- ○ Prefer not to say (4)

*

What is your age?

____

What ethnicity would you describe yourself as?

- ☐ Chinese (1)
- ☐ Malay (2)
- ☐ Indian (3)
- ☐ Eurasian (7)
- ☐ Other (6) ____
- ☐ Prefer not to say (8)

What is the highest degree or level of school you have completed?

▼ Less than a high school diploma (1) ... Doctorate or professional degree (7)

How familiar are you with robots?

	Not Familiar (1)	Slightly Familiar (2)	Moderately Familiar (3)	Very Familiar (4)	Extremely Familiar (5)
Familiarity (1)	○	○	○	○	○

How comfortable are you in working with robots?

	Not Comfortable (1)	Slightly Comfortable (3)	Moderately Comfortable (4)	Very Comfortable (5)	Extremely Comfortable (7)
Comfort (1)	○	○	○	○	○

Page Break

Section 2: Analyzing Robot Expressive Movements and Mental State In this section, you will see a set of videos showing different expressive movements of robots.

Observe each expression closely

(Optional) For each expression, what words would you use to describe the mental state of the robot?

- Expression A (1) ____
- Expression B (2) ____
- Expression C (3) ____
- Expression D (4) ____

JS

Select the mental state of the robot that best matches **Expression A**

▼ Confidence (1) ... None (8)

Display this question:

If Select the mental state of the robot that best matches Expression A = Other

If other, then please enter the mental state that best matches **Expression A**

____

Display this question:

If Select the mental state of the robot that best matches Expression A = Confidence

Or Select the mental state of the robot that best matches Expression A = Curiosity

Or Select the mental state of the robot that best matches Expression A = Fear

Or Select the mental state of the robot that best matches Expression A = Hesitation

Or Or If other, then please enter the mental state that best matches Expression A Text Response Is Not Empty

JS

Rate the degree of **(no selection yet)** in Expression A, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

○ 1 (1)

○ 2 (4)

○ 3 (5)

○ 4 (6)

○ 5 (7)

JS

Select the mental state of the robot that best matches **Expression B**

▼ Confidence (1) ... None (8)

Display this question:

If Select the mental state of the robot that best matches Expression B = Other

If other, then please enter the mental state that best matches **Expression B**

____

Display this question:

If Select the mental state of the robot that best matches Expression B = Confidence

Or Select the mental state of the robot that best matches Expression B = Curiosity

Or Select the mental state of the robot that best matches Expression B = Fear

Or Select the mental state of the robot that best matches Expression B = Hesitation

Or Or If other, then please enter the mental state that best matches Expression B Text Response Is Not Empty

JS

Rate the degree of **(no selection yet)** in Expression B, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

○ 1 (1)

○ 2 (4)

○ 3 (5)

○ 4 (6)

○ 5 (7)

JS

Select the mental state of the robot that best matches **Expression C**

▼ Confidence (1) ... None (8)

Display this question:

If Select the mental state of the robot that best matches Expression C = Other

If other, then please enter the mental state that best matches **Expression C**

____

Display this question:

If Select the mental state of the robot that best matches Expression C = Confidence

Or Select the mental state of the robot that best matches Expression C = Curiosity

Or Select the mental state of the robot that best matches Expression C = Fear

Or Select the mental state of the robot that best matches Expression C = Hesitation

Or Or If other, then please enter the mental state that best matches Expression C Text Response Is Not Empty

JS

Rate the degree of **(no selection yet)** in Expression C, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

○ 1 (1)

○ 2 (4)

○ 3 (5)

○ 4 (6)

○ 5 (7)

JS

Select the mental state of the robot that best matches **Expression D**

▼ Confidence (1) ... None (8)

Display this question:

If Select the mental state of the robot that best matches Expression D = Other

If other, then please enter the mental state that best matches **Expression D**

____

Display this question:

If Select the mental state of the robot that best matches Expression D = Confidence

Or Select the mental state of the robot that best matches Expression D = Curiosity

Or Select the mental state of the robot that best matches Expression D = Fear

Or Select the mental state of the robot that best matches Expression D = Hesitation

Or Or If other, then please enter the mental state that best matches Expression D Text Response Is Not Empty

JS

Rate the degree of **(no selection yet)** in Expression D, with 1 being slightly (no selection yet) and 5 being extremely (no selection yet).

- ○ 1 (1)
- ○ 2 (4)
- ○ 3 (5)
- ○ 4 (6)
- ○ 5 (7)

End of Block: Background Information

Start of Block: Block 5

In the next set of questions, you'll see slight variations of each robot expression. Choose the robot that shows the mental state more strongly.

End of Block: Block 5

Start of Block: Compare: Confidence

Which robot appears more **confident**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally confident (4)
- Neither appears confident (5)

Which robot appears more **confident**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally confident (4)
- Neither appears confident (5)

End of Block: Compare: Confidence

Start of Block: Compare: Fear

Which robot appears more **fearful**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally fearful (4)
- Neither appears fearful (5)

Which robot appears more **fearful?**

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally fearful (4)
- Neither appears fearful (5)

Which robot appears more **fearful**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally fearful (4)
- Neither appears fearful (5)

Which robot appears more **fearful**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally fearful (4)
- Neither appears fearful (5)

Which robot appears more **fearful**?

- ○ Robot 1 (1)
- ○ Robot 2 (2)
- ○ Both appear equally fearful (4)
- ○ Neither appears fearful (5)

Which robot appears more **fearful**?

- ○ Robot 1 (1)
- ○ Robot 2 (2)
- ○ Both appear equally fearful (4)
- ○ Neither appears fearful (5)

End of Block: Compare: Fear

Start of Block: check

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Please enter your birth year

____

End of Block: check

Start of Block: Compare: Curious

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

Which robot appears more **curious**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally curious (4)
- Neither appears curious (5)

End of Block: Compare: Curious

Start of Block: Compare: Hesitant

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appears equally hesitant (4)
- Neither appears hesitant (5)

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally hesitant (4)
- Neither appears hesitant (5)

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally hesitant (4)
- Neither appears hesitant (5)

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally hesitant (4)
- Neither appears hesitant (5)

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally hesitant (4)
- Neither appears hesitant (5)

Which robot appears more **hesitant**?

- Robot 1 (1)
- Robot 2 (2)
- Both appear equally hesitant (4)
- Neither appears hesitant (5)

End of Block: Compare: Hesitant
